IT3218- Artificial Intelligence

What is Artificial Intelligence?

- **Artificial intelligence is the technology that allows systems to replicate human behavior and thoughts.**

- At its core, AI uses algorithms to train datasets that will generate AI models that let computer systems perform tasks like recommending songs, googling route directions, or providing text translations betweentwo languages.

- A few examples of AI are ChatGPT, **Google Translate, Tesla, Netflix,** and many more.

- ***According to the father of artificial intelligence, John McCarthy, it is The science and engineering of making intelligent machines, especially intelligent computer programs..***

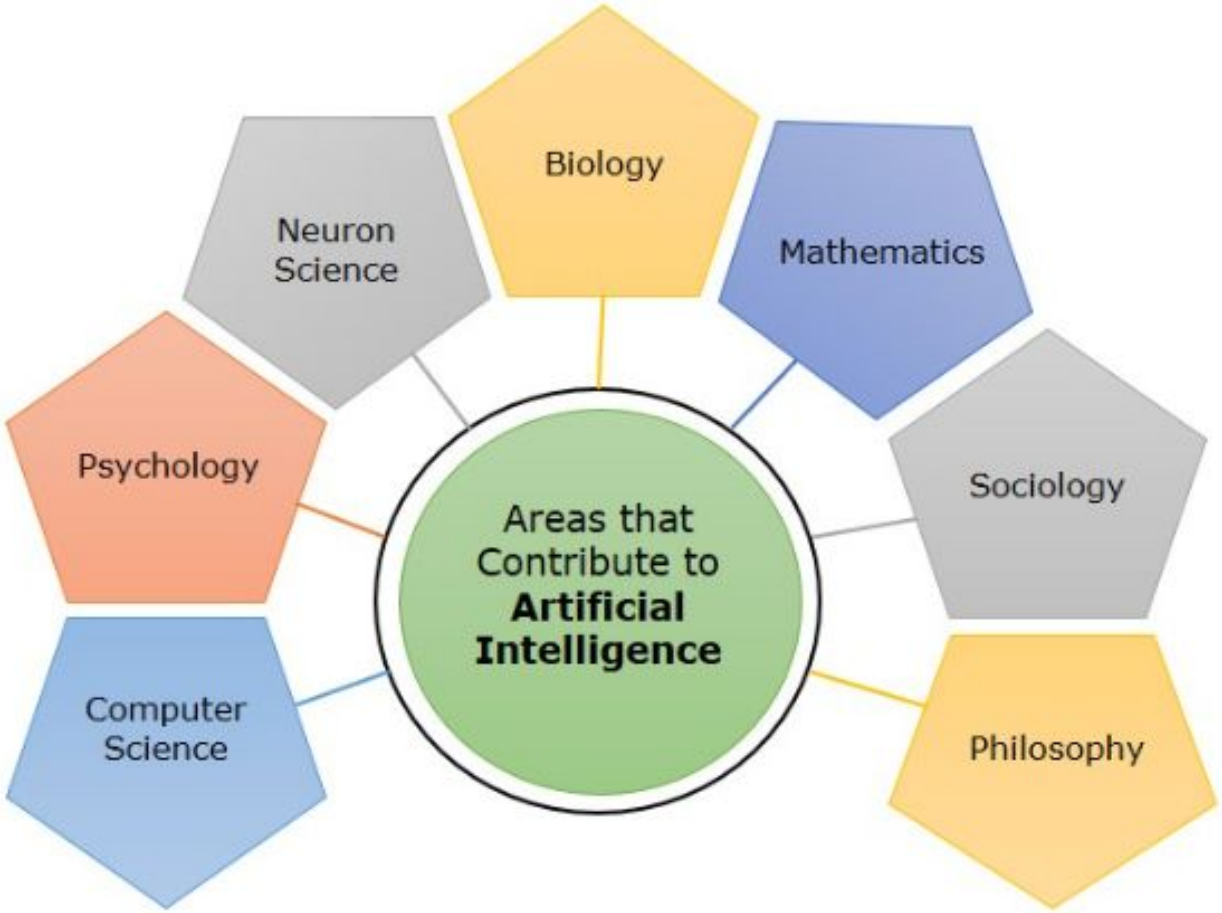
History of AI

Year	Milestone
1923	**Karel apek** play named Rossum's Universal Robots (RUR) opens in London, first use of the word **"robot"** in English.
1956	**John McCarthy**, a professor at Dartmouth College coined the term "Artificial Intelligence".
1966	Joseph Weizenbaum created **ELIZA**, that used **natural language processing** to make conversations with humans.
1997	**Deep Blue** was the first program to beat a human chess champion, Gray Kasparov.
2012	**AlexNet** is a convolution neural network (CNN) architecture that was designed by Alex Krizhevsky.
2020	OpenAI started beta testing **GPT-3**, a model that uses deep learning to create code, content, and other creative tasks.

Goals of AI

- To Create Expert Systems
- To Implement Human Intelligence in Machines
- To Develop Problem-Solving Ability
- To Allow Continuous Learning
- To Encourage Social Intelligence and Creativity

What Contributes to AI?



AI Programming vs. Traditional Coding

AI Programming	Traditional Coding
Can deal with complex, undefined problems.	Can handle only well-defined, predictable problems.
Uses data-driven methods and algorithms.	Relies on explicit logic and rules.
Produces models that make predictions or decisions.	Generates specific functional software
Utilizes frameworks and libraries like **TensorFlow**, **PyTorch**.	Commonly uses languages like **Python**, **Java**.
Involves validation of model accuracy.	Focuses on debugging and unit testing.
Models learn patterns from data.	Programs execute pre-defined instructions.

What is an AI Technique?

- **AI techniques** refer to methods and algorithms that are used to create smart systems that perform tasks requiring human-like intelligence. Some of these techniques are **Machine Learning**, **Natural Language Processing**, **Computer Vision** and others. These AI techniques use the knowledge efficiently in such a way that –

1) It should be perceivable by the people who provide it.
2) It should be easily modifiable to correct errors.
3) Elevate the speed of execution of the complex program it is equipped with.

Applications of AI

- **Gaming** − AI plays a crucial role in strategic games such as **chess, poker, tic-tac-toe,** etc., where machine can think of large number of possible positions based on heuristic knowledge.
- **Natural Language Processing** − It enables machines to interact with humans in natural language.
- **Expert Systems** − It is an AI based software that enables decision-making ability similar to a human expert.
- **Computer Vision** − These systems understand, interpret, and comprehend visual input on the computer.

Applications of AI

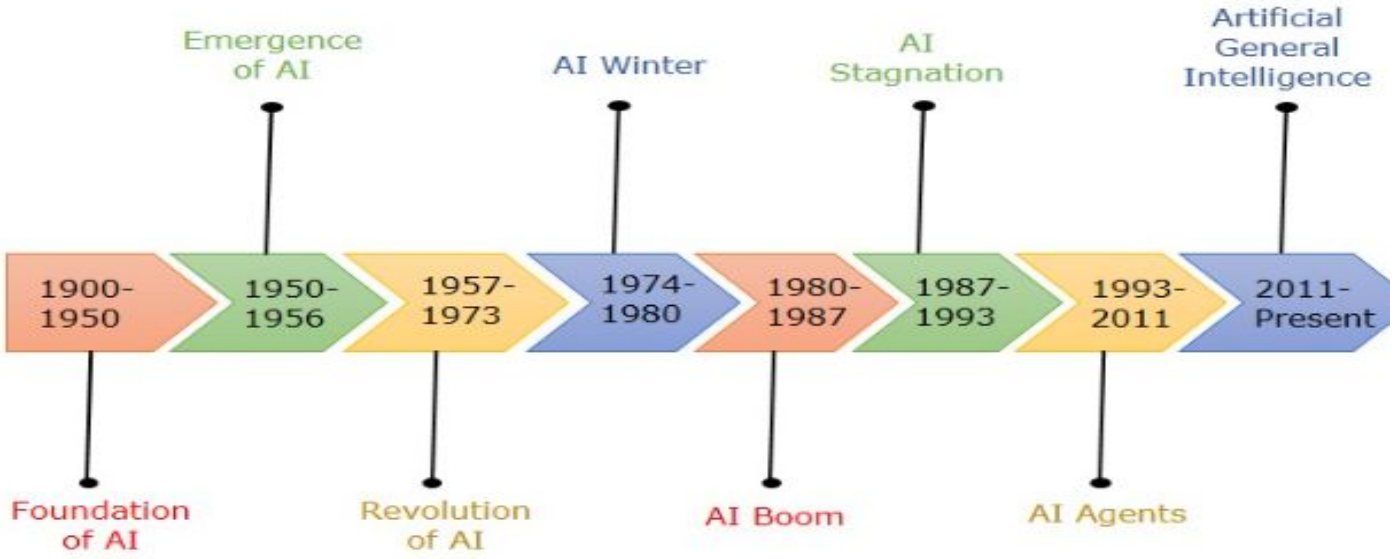
- **Speech Recognition** − Some intelligent systems are capable of hearing and comprehending the language in terms of sentences and their meanings while a human talks to it. It can handle different accents, slang words, noise in the background, change in humans noise due to cold, etc.
- **Handwriting Recognition** − The handwriting recognition software reads the text written on paper by a pen or on screen by a stylus. It can recognize the shapes of the letters and convert it into editable text.
- **Intelligent Robots** − Robots are able to perform the tasks given by a human. They have sensors to detect physical data from the real world such as temperature, movement, and sound. They have efficient processors, and huge memory, to exhibit intelligence. In addition, they are capable of learning from their mistakes and they can adapt to new environment.

Challenges in AI

- **Data Quality and Accessibility** − AI requires large, high-quality, and relevant datasets for effective learning.
- **Technical Expertise** − Implementing AI algorithms and models requires skilled professionals.
- **Ethical and Legal Concerns** − It is important to make sure that the AI systems are fair, unbiased, and don't harm anyone's safety.
- **Integration** − Integrating AI with existing systems can be complex.
- **Cost** − Developing and maintaining of AI infrastructure can be expensive.

History & Evolution

- *There is an assumption that **Artificial Intelligence** is a recent technology in the market, but in reality the groundwork of AI dates back to early 1900s, while the biggest innovations weren't made until 1950's.*



Evolution Artificial Intelligence

Evolution -Artificial Intelligence

- Foundation of AI

Year	Milestone
1921	Czech playwright Karel Capek released a science fiction play **"Rossum's Universal Robots"**, where he introduced artificial people and named them robots.
1943	Warren McCulloch and Walter Pitts created the first conceptual model of a **neural network**.

Year	Milestone
1950	Alan Turing published **"Computer Machinery and Intelligence"** which proposed Turing test to measure the intelligence of a machine.
1952	**Arthur Samuel** is a computer scientist, who developed a program to play checkers, which improved its performance through experience.

Evolution -Artificial Intelligence

- AI Revolution

Year	Milestone
1957	**Frank Rosenblatt** introduced **perceptron**, which was one of the early innovation for **artificial neural networks**.
1958	John McCarthy created **LISP**, the first programming language for AI research.
1959	Arthur Samuel used the term **"Machine Learning"** and defined it as intellectual computers that surpass humans in any task.
1966	Joseph Weizenbaum created **ELIZA**, that used **natural language processing** to make conversations with humans.
1972	Alain Colmerauer and Philippe Roussel developed **prolog programming language.**

Evolution -Artificial Intelligence

- AI Winter

The initial AI winter occurred from **1974-1980** , which was quite a tough time for the improvement of AI. During this time, there was a substantial decrease in research funding which affected the interest on AI.

- AI Boom

Year	Milestone
1980	The first **expert systems**, know as **XCON** came into the commercial market.
1981	The Japanese government alocated $850 million to the development of **Fifth Generation Computer Project**, to create computers that could translate, converse in human language and express reasoning on a human level.
1984	The AAAI warns about the incoming **AI Winter**, where the funding and interest would decrease significantly effecting the research.
1986	Ernst Dickmann and his team demonstrated the first self-driving cars, which drove up to 55kmph with no obstacles and human driver.

Evolution- Artificial Intelligence

- AI Stagnation

The second AI winter took place from the years **1987-1993**, where again investors and government stopped funding due to high cost and no efficient results.

- AI Agents

Year	Milestone
1997	**Deep Blue** was the first program to beat a human chess champion, Gray Kasparov.
2000	Professor Cynthia Breazeal the first robot named **Kismet**that could simulate human emotions and had facial features similar to humans.
2003	NASA landed two rovers onto Mars, which navigated through the surface of the planet without human intervention.
2006	Companies such as Twitter, Facebook, and Netflix started using AI as part of advertising, business analysis, and user engagement.
2011	Apple released Siri, the first popular voice assistant.

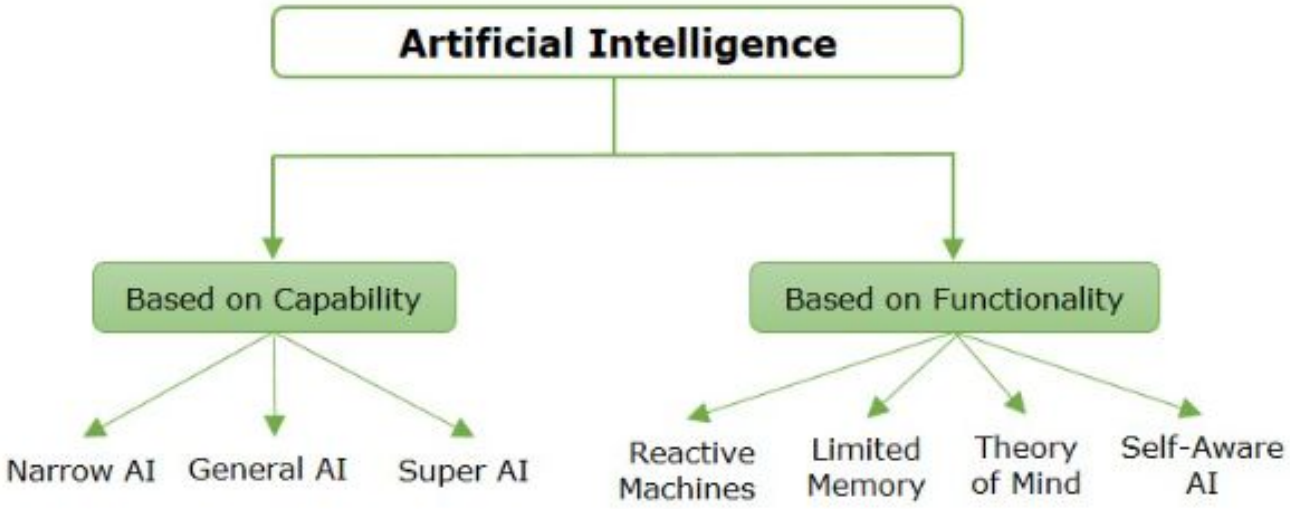
Evolution -Artificial Intelligence

- Artificial General Intelligence

Year	Milestone
2012	Google researchers Jeff Dean and Andrew Ng trained a neural network to recognize cats using unlabeled images without prior information.
2016	Hanson Robotics introduced **Sophia**, the first humanoid robot with realistic human features, emotion recognition, and communication abilities.
2017	Facebook programmed two AI Chatbot to communicate and learn to negotiate, but as the conversation went on they eventually stopped using English and started speaking their own language entirely on their own.
2018	A Chinese tech group **Alibaba's language – processing AI** won over human intellect on a Standford reading and comprehension test.
2019	**Google's AlphaStar** reached Grandmaster on the video game StarCraft 2 outperforming all but .2% of human players.
2020	OpenAI started beta testing **GPT-3**, a model that uses Deep Learning to create code, content, and other creative tasks.
2021	OpenAI developed **Dall-E**, which can generate images using the natural language as prompts.
2022	Dall-E was integrated with **ChatGPT**, showcasing AI's capacity to generate text and relevant images.
2023	**Multimodal** is another major breakthrough in AI. These models process all the data types like text, image, video, and audio simultaneously.
2024	**Devin** is the first AI software engineer still under development and **SORA** is another innovation of OpenAI which is an text-to-video model.

Artificial Intelligence - Types

- **Artificial Intelligence (AI)** is a technology that enables computers to think and act like humans. These systems are trained to learn from past experiences to enhance speed, precision, and effectiveness. Further based on the following criteria artificial intelligence can be categorized into different types −



Artificial Intelligence - Terminology

Term	Definition
Artificial Intelligence (AI)	The technology that enable computers and machine to replicate human intelligence.
Machine Learning (ML)	A subset of AI that allows systems to learn from data and improve their performance over time.
Deep Learning	A specialized domain of machine learning that uses neural networks with many layers to analyze various forms of data.
Neural Networks	Computational models inspired by the functioning of human brain using neurons. This models consist of interconnected nodes to process the data.
Natural Language Processing (NLP)	The domain in AI which deals with interaction between computers and humans through natural language.
Computer Vision	A field of AI that allows machines to interpret and make decisions over visual data.

Artificial Intelligence - Terminology

Reinforcement Learning	This is a type of machine learning in which an agent learns to make decisions based on actions in an environment.
Supervised Learning	A type of machine learning where the model is trained on labeled data to predict outcomes.
Unsupervised Learning	A type of machine learning where the model identifies patters and relationships from unlabeled data.
Semi-Supervised Learning	A hybrid machine learning method that combines small amount of labeled data and a large amount of unlabeled data to predict outcomes.
Data Mining	The process of discovering patterns and knowledge from large amounts of data using various techniques.
Agent	An entity that perceives its environment and takes actions to achieve specific goals.
Algorithm	A step-wise procedure or processes followed in calculations or problem-solving operations by a computer.
Training Data	The dataset used to train a machine learning model to recognize patterns and make predictions.
Model	A mathematical representation of a process which captures relationships in the data for predictive tasks.

Artificial Intelligence - Terminology

Overfitting	A modeling error that occurs when a model learns the training data too well, capturing noise instead of the underlying patterns.
Underfitting	A modeling error that occurs when a model is too simple to capture the underlying trend in the data.
Cognitive Computing	An AI approach that mimics human through processes in a complex, human-like way.
Autonomous	Systems that operate independently without human intervention.
Large Language Models	AI models like GPT that are trained on large amounts of text data to understand and generate human-like data.
Artificial General Intelligence (AGI)	A theoretical form of AI that is capable of understanding, learning, and general intelligence throughout almost any task, quite similar to that of a human being.
Generative AI	AI capable of generating new content, be it text, images, or music, based on learned patterns.

Artificial Intelligence - Terminology

Transfer Learning	A technique where the model trained on one task is adapted to work on a different related tasks.
Chatbot	A program designed to simulate conversation with human users.
Backward Chaining	An inference method where reasoning started from the goal and works backwards to find supporting data.
Forward Chaining	An inference method that started with available data and applied rules to extract more data until a goal is reached.
Environment	The surrounding context or scenario in which an agent operates and makes decisions.
Heuristics	Problem-solving strategies that use practical methods to produce solutions that may not be optimal but are sufficient for immediate goals.

Artificial Intelligence - Tools & Frameworks

- PyTorch
- Scikit-Learn
- TensorFlow
- Keras
- Microsoft Cognitive Toolkit(CNTK)
- LangChain
- Hugging Face
- PyBrain
- Theano
- Caffe
- XGBoost